

MODUL PRAKTIKUM
REKAYASA PERANGKAT LUNAK



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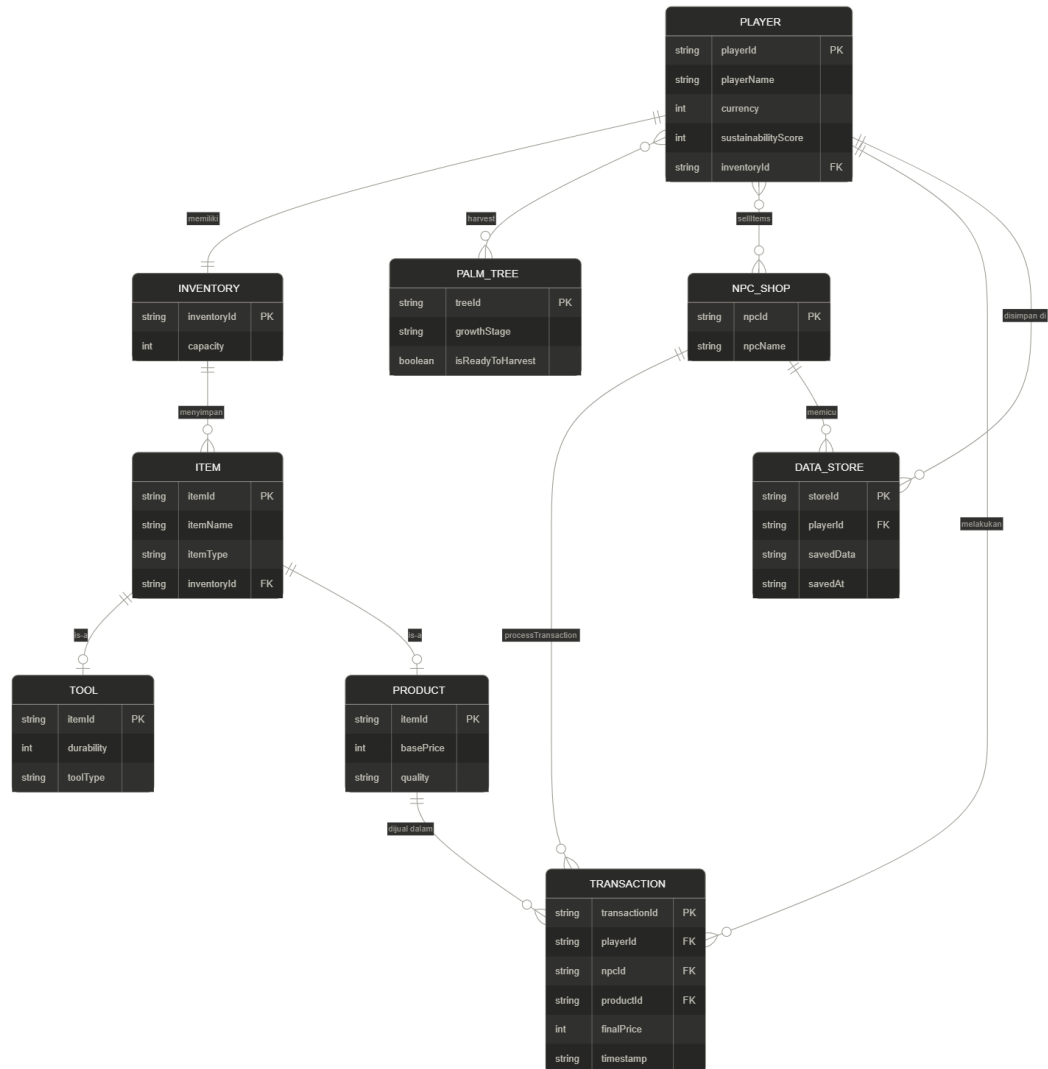
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Nama Proyek : Sawit : The Last Chance

PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS TEKNOLOGI INDUSTRI DAN INFORMATIKA
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Pertemuan 9 – Perancangan Basis Data

Entity Relationship Diagram (ERD)



Tabel & Atribut

PLAYER

- playerId (PK) — String
- playerName — String
- currency — Integer
- sustainabilityScore — Integer
- inventoryId (FK → INVENTORY) — String

INVENTORY

- inventoryId (PK) — String
- capacity — Integer

ITEM (tabel parent / supertype)

- itemId (PK) — String
- itemName — String
- itemType — String (*nilai: "tool" atau "product"*)
- inventoryId (FK → INVENTORY) — String

TOOL (*subtype dari ITEM*)

- itemId (PK, FK → ITEM) — String
- durability — Integer
- toolType — String

PRODUCT (*subtype dari ITEM*)

- itemId (PK, FK → ITEM) — String
- basePrice — Integer
- quality — String

PALM_TREE

- treeId (PK) — String
- growthStage — String
- isReadyToHarvest — Boolean

NPC_SHOP

- npcId (PK) — String
- npcName — String

TRANSACTION (*tabel junction*)

- transactionId (PK) — String
- playerId (FK → PLAYER) — String
- npcId (FK → NPC_SHOP) — String
- productId (FK → PRODUCT) — String
- finalPrice — Integer
- timestamp — String

PLAYER_HARVEST (*tabel junction many-to-many*)

- playerId (FK → PLAYER) — String
- treeId (FK → PALM_TREE) — String
- harvestedAt — String

DATA_STORE

- storeId (PK) — String
- playerId (FK → PLAYER) — String
- savedData — String
- savedAt — String

Relasi

Relasi	Tabel Asal	Tabel Tujuan	Kardinalitas	Keterangan
memiliki	PLAYER	INVENTORY	1 : 1	Satu player punya tepat satu inventory
menyimpan	INVENTORY	ITEM	1 : N	Satu inventory menyimpan banyak item
is-a (inheritance)	ITEM	TOOL	1 : 0..1	Item bisa berupa tool
is-a (inheritance)	ITEM	PRODUCT	1 : 0..1	Item bisa berupa product

Relasi	Tabel Asal	Tabel Tujuan	Kardinalitas	Keterangan
harvest	PLAYER	PALM_TREE	M : N	Lewat tabel junction PLAYER_HARVEST
sellItems / bertransaksi	PLAYER	NPC_SHOP	M : N	Lewat tabel TRANSACTION
processTransaction	NPC_SHOP	TRANSACTION	1 : N	Satu NPC memproses banyak transaksi
dijual dalam	PRODUCT	TRANSACTION	1 : N	Satu produk bisa muncul di banyak transaksi
memicu / menyimpan	NPC_SHOP	DATA_STORE	1 : N	Transaksi memicu penyimpanan data
disimpan di	PLAYER	DATA_STORE	1 : N	Data satu player bisa disimpan berkali-kali

Class Diagram

